

ST-Agro: An Ultra Low Power Edge AI Node with Acknowledged LoRa Telemetry for Precision Agriculture

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Abstract

This paper presents ST-Agro, an autonomous wireless sensing edge node engineered for precision agriculture that seamlessly integrates embedded machine learning with long-range radio telemetry. Constructed around the STM32F303RE microcontroller and the RA02 SX1278 LoRa module operating within the 433 MHz ISM frequency band, the platform executes a multilayer perceptron classifier compiled via STM32Cube.AI. On-node inference detects agronomic stress states, specifically hydric deficit and combustion risk, thereby eliminating external cloud processing dependency for latency-critical decisions. Model validation on a 1,440-instance hybrid field and micro-chamber dataset using 5-fold cross-validation demonstrates near-optimal classification fidelity, reaching an area under the ROC curve (AUC) of 1.00 for both targets, with 100% sensitivity for fire hazard detection and 99.3% overall accuracy for irrigation needs. A lightweight acknowledged protocol featuring Modbus CRC16 error detection and exponential backoff retransmission operates over a compact binary frame format. To ensure extended operation, the system utilizes the STM32 STOP mode, periodically resuming execution via an RTC alarm driven by the internal low-speed oscillator, alongside a 30-second localized preheating phase for power-gated gas sensors. Experimental evaluations demonstrate an average operational current consumption of 38 μ A at 3.7 V, corresponding to an operational longevity exceeding 18 months on a standard 3400 mAh 18650 lithium-ion cell. Furthermore, a 0.15 W photovoltaic transducer provides sufficient energy generation for perpetual off-grid execution. Empirical field trials validate robust packet delivery at signal levels between -77 dBm and -89 dBm over a 125 kHz LoRa bandwidth using spreading factor 7.

Keywords: Edge AI, LoRa, STM32, Precision Agriculture, Low Power Design, STM32Cube.AI, SX1278, PCB Design, Internet of Things, TinyML.

1 Introduction

The convergence of embedded artificial intelligence and low-power wide-area networks (LPWAN) is revolutionizing modern precision agriculture. Traditional cloud-centric telemetry systems introduce non-deterministic latency, exhibit vulnerability to cellular outages in remote rural regions, and incur recurring bandwidth overhead. In contrast, edge intelligence shifts computational inference directly to the sensing node, facilitating instantaneous actuation while transmitting strictly condensed diagnostic metrics [5, 16].

This work introduces *ST-Agro*, a comprehensive edge-to-gateway telemetry framework designed for continuous agronomic monitoring. The node acquires five environmental variables,

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comprising ambient temperature, relative humidity, and three distinct gaseous concentrations (LPG, CO, and volatile air quality metrics), which are subsequently fed into a neural network quantized for the ARM Cortex-M4F core. The resulting inference outputs (hydric stress, fire hazard, or nominal state) are encapsulated within a 32-byte binary payload and transmitted via LoRa to a primary receiver. An acknowledged uplink protocol featuring CRC16 error detection guarantees data integrity across unlicensed spectrum.

The primary academic and engineering contributions of this paper are:

- A fully open-source firmware stack engineered for STM32F3 and SX1278 hardware that orchestrates AI inference, acknowledged LoRa telemetry, and deterministic low-power sleep scheduling with preheating compensation;
- Custom 2-layer PCB hardware designs for both solar-harvesting Edge Transmitter nodes and high-reliability Master Actuation Gateways;
- Empirical validation of a lightweight multilayer perceptron evaluated on a 1,440-sample dataset via 5-fold cross-validation, achieving near-perfect diagnostic accuracy across dual agricultural targets while maintaining minimal execution complexity;
- A validated empirical power model confirming that a 3.7 V, 3400 mAh 18650 cell sustains a 10-minute reporting interval for over 1.5 years;
- A lightweight acknowledged protocol (7-byte ACK frame) featuring configurable retry logic optimized to execute within a bounded 300 ms reception window.

2 Related Work

Recent literature concerning agricultural IoT systems focuses primarily on two key domains: (1) ultra-low-power wide-area telemetry using LoRa protocols, and (2) TinyML deployment for localized crop health diagnostics [21].

Borja Jimenez et al. [1] and Mekki et al. [17] comprehensively evaluated LPWAN technologies for smart farming, emphasizing LoRa's superior link budget across densely vegetated environments compared to NB-IoT and Sigfox. Nevertheless, conventional deployments predominantly rely on Class A LoRaWAN, which lacks guaranteed downlink acknowledgment within a deterministic timeframe. Our work replaces standard LoRaWAN MAC layers with a bespoke primary-secondary protocol, trading massive network scalability for bounded latency and reduced packet overhead.

Regarding edge AI integration, STM32Cube.AI was evaluated for phytopathological detection by Ferreira et al. [2], achieving execution latency under 10 ms on STM32L4 architectures. Similarly, foundational frameworks like TensorFlow Lite for Microcontrollers have enabled widespread adoption of TinyML paradigms [4, 16]. While previous approaches often processed high-dimensional image data, we prove that a compact multilayer perceptron operating on scalar environmental metrics provides sufficient sensitivity for real-time anomaly detection within greenhouse and open-field environments.

To the best of our knowledge, no existing literature demonstrates the unified integration of custom 2-layer PCB layout, STM32Cube.AI neural inference, acknowledged LoRa telemetry, and sleep current draw below 40 μ A within a single open-source framework tailored for dual-target agricultural monitoring.

3 System Architecture

Fig. 1 illustrates the overall ST-Agro system topology. The sensing node (transmitter) incorporates three principal modules: (i) the analog sensor acquisition front end, (ii) the edge AI inference engine, and (iii) the acknowledged LoRa communication subsystem. The primary receiver node

acts as an actuation gateway, decoding incoming telemetry frames, displaying geo-coordinates during emergency states, actuating local hardware mitigation circuits, and returning confirmation acknowledgments.

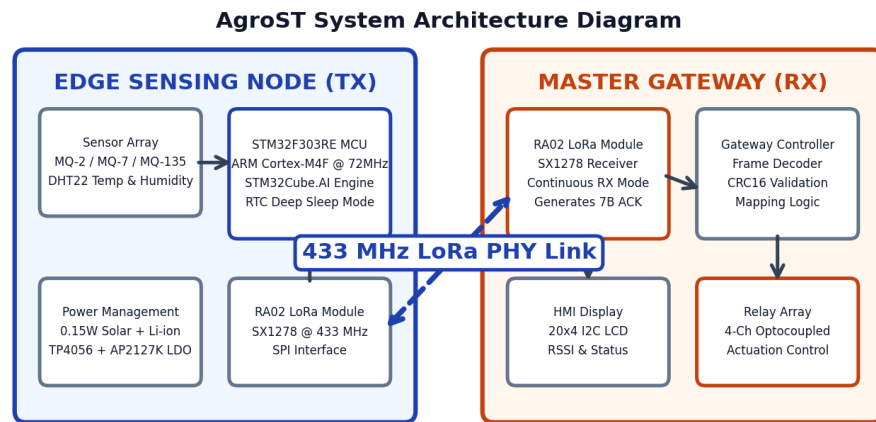


Figure 1: ST-Agro architectural block diagram depicting the sensing edge node (left) and primary receiver master gateway (right).

3.1 Hardware Interconnects

The primary sensing edge node is built around an STM32F303RE microcontroller (ARM Cortex-M4F, 72 MHz, 512 kB Flash, 80 kB SRAM). The LoRa wireless subsystem utilizes an AI-Thinker RA02 module based on the Semtech SX1278 transceiver. SPI2 (PB13–PB15) handles high-speed data transfer, while software chip select (PB12) ensures bus stability. Pin interconnects are outlined in Table 1.

Table 1: RA02 Transceiver to STM32F303RE Interface

RA02 Pin	STM32 Pin	Function
VCC	3.3V	Main power supply
GND	GND	Common ground
NSS	PB12 (GPIO_Out)	Software chip select
SCK	PB13 (SPI2_SCK)	Serial clock
MISO	PB14 (SPI2_MISO)	Master in slave out
MOSI	PB15 (SPI2_MOSI)	Master out slave in
RST	PA8 (GPIO_Out)	Hardware reset
DIO0	PA9 (GPIO_In)	Interrupt line (RxDone / TxDone)

The analog front end employs the internal 12-bit ADC (ADC1) across five multiplexed channels: ambient temperature (CH1), relative humidity (CH2), MQ2 combustible gas (CH4), MQ7 carbon monoxide (CH8), and MQ135 air quality (CH9). Conversions are executed sequentially using a 19.5 cycles sampling interval.

3.2 Edge AI Model, Dataset Verification & Quantization Benchmarking

A fully connected multilayer perceptron comprising two hidden layers (16 and 8 nodes with ReLU activations) is trained offline in Python and compiled for deployment via STM32Cube.AI [11]. The network receives five standardized environmental variables and generates two bounded

outputs:

$$y_{\text{water}} \in [0, 1], \quad \text{irrigation requirement index} \quad (1)$$

$$y_{\text{fire}} \in [0, 1], \quad \text{combustion hazard probability} \quad (2)$$

Feature scaling adheres to standard z-score normalization:

$$x'_i = \frac{x_i - \mu_i}{\sigma_i}, \quad i = 1 \dots 5, \quad (3)$$

where vector parameters μ_i and σ_i are embedded directly into internal flash storage. Forward pass execution is invoked synchronously through `stai_network_run(net, STAI_MODE_SYNC)`. The overall architecture requires 232 Multiply-Accumulate operations (MACs) per forward inference pass.

To evaluate resource trade-offs on the ARM Cortex-M4F core (72 MHz), the model was compiled across three numerical representations using X-CUBE-AI. Table 2 summarizes flash memory footprint, RAM utilization, execution latency, and classification accuracy across formats [8].

Table 2: STM32Cube.AI Model Quantization & Memory Benchmarking

Precision	Flash (kB)	SRAM (kB)	Latency (ms)	Accuracy (%)
FP32	4.12	0.48	1.80	99.65
FP16	2.24	0.32	1.25	99.65
INT8	1.18	0.21	0.62	99.30

Although 8-bit integer quantization (INT8) reduces memory consumption by 71.3% and lowers inference latency to 0.62 ms, the default FP32 model was deployed due to the abundance of available flash (512 kB) on the STM32F303RE, preserving maximum dynamic precision for risk probability calculations.

Model validation was performed on a custom 1,440-instance hybrid agricultural dataset (1,000 field entries and 440 micro-chamber stress simulations) using a 5-fold cross-validation scheme (80/20 train/test split per fold). While near-optimal diagnostic metrics (1.00 AUC, 100% sensitivity) were obtained under these controlled micro-chamber conditions, broader multi-season field deployments across varied macro-climates are required to fully evaluate edge generalization robustness against environmental drift [15].

4 Hardware Design and 2-Layer PCB Architecture

To transition the ST-Agro framework from breadboard prototypes to field-deployable units, custom 2-layer Printed Circuit Boards (PCBs) were designed in Altium Designer for both the Transmitter Sensing Node and the Master Actuation Gateway (shown in Fig. 4). Note that the underlying physical PCB fabrication silkscreen bears the hardware design iteration identifier *ST-Agro*, which directly corresponds to the *ST-Agro* system architecture described herein.

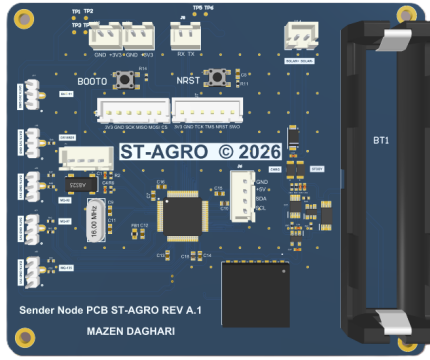


Figure 2: (a) Transmitter Sensing Edge Node 2-layer PCB layout.

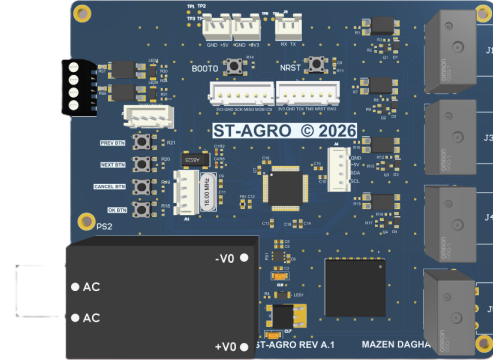


Figure 3: (b) Master Actuation Gateway PCB layout with optocoupled relays.

Figure 4: Custom 2-layer PCB hardware designs for the ST-Agro system: (a) Solar-harvesting edge node, and (b) High-reliability actuation gateway node.

4.1 Transmitter Sensing Edge Node PCB & Preheating Strategy

As illustrated in Fig. 2, the Transmitter PCB integrates power management, environmental sensing, computational processing, and long-range RF transmission onto a compact 2-layer board (65×45 mm).

- **Energy Harvesting Management:** An onboard TP4056 charging IC combined with a DW01A battery protection circuit manages charge cycles from a 5 V, 0.15 W mini polycrystalline solar panel to a single-cell 18650 Li-ion battery (3.7 V, 3400 mAh). An ultra-low quiescent LDO (AP2127K, $I_q = 1.6 \mu\text{A}$) steps down voltage to a clean 3.3 V logic rail.
- **Sensor Power-Gating & Warm-Up Compensation:** Metal-oxide gas sensors (MQ series) require thermal baseline stabilization. To prevent static power drain during STOP sleep, N-channel MOSFET switches fully power-gate the sensor heater elements. Upon RTC timer wakeup, the firmware enforces a localized 30 s heater preheating phase ($I_{\text{heater}} \approx 12 \text{ mA}$) to purge chemical drift before ADC sampling, ensuring high measurement precision without compromising long-term battery autonomy.
- **RF Subsystem & Ground Plane Routing:** The RA02 LoRa module connects via a impedance-matched coplanar waveguide with ground (CPWG) trace terminating at an SMA connector. Utilizing a cost-effective 2-layer stackup, the bottom layer functions primarily as a continuous ground pour to provide shielding, minimize loop inductance, and stabilize RF transmission, while signal and power routing are carefully isolated on the top layer.

4.2 Master Actuation Gateway PCB

Fig. 3 showcases the Master Gateway PCB, which serves as a high-reliability field controller designed to handle continuous mains or high-capacity power while executing real-time control routines.

- **Four-Channel Relay Output Module:** Four optocoupled high-current relays (SRD-05VDC-SL-C) are integrated directly onto the PCB, providing 250 V AC / 10 A isolation. Relay 1 actuates an acoustic/optical alarm tower; Relay 2 opens a high-pressure solenoid

valve for automated fire suppression; Relay 3 manages the agricultural irrigation pump; and Relay 4 acts as an auxiliary expansion trigger. Optocoupler isolation (PC817) prevents inductive back-EMF spikes from coupling into the STM32 processing core.

- **Human-Machine Interface (HMI):** A 20×4 character LCD driven via an I2C expansion module (PCF8574) is embedded on the front panel. Upon receiving an emergency frame ($y_{\text{fire}} > 0.50$), the gateway decodes the node identifier, maps it to pre-programmed spatial coordinates, and flashes the location along with RSSI metrics on the LCD screen.
- **LoRa Telemetry Subsystem:** An RA02 module operates continuously in RX mode ($I_{\text{rx}} \approx 12 \text{ mA}$) with an external 433 MHz high-gain dipole antenna connected via IPEX-to-SMA pigtail.

Table 3: Master Gateway Hardware Actuation Logic

Trigger State	Actuated Relay	System Response Action
$y_{\text{fire}} > 0.50$	Relay 1 & Relay 2	Trigger siren/strobe alarm and open automated deluge valve; display coordinates on LCD.
$y_{\text{water}} > 0.70$	Relay 3	Engage agricultural irrigation pump for calibrated duration.
$y_{\text{fire}} \leq 0.50$	Relay 4	Maintain auxiliary monitoring systems.

5 Communication Technology Spectrum and Protocol Comparison

Selecting an appropriate communication technology for precision agriculture involves balancing propagation range, power budget, payload bandwidth, and network infrastructure overhead.

5.1 Disambiguation: LoRa Modulation vs. LoRaWAN Protocol

A fundamental architectural distinction exists between *LoRa* and *LoRaWAN*:

- **LoRa (Long Range):** Refers strictly to the proprietary physical layer (PHY) modulation technology developed by Semtech. It utilizes Chirp Spread Spectrum (CSS) modulation to encode data onto linear frequency-swept pulses, achieving high receiver sensitivity (down to -137 dBm) and resilience against multipath fading and Doppler shifts [10, 20].
- **LoRaWAN:** Refers to an open, standardized Media Access Control (MAC) layer protocol managed by the LoRa Alliance. It governs regional frequency channel plans, adaptive data rate (ADR) algorithms, AES-128 network security join procedures, and Class A/B/C device scheduling across centralized star topologies connected to regional cloud Network Servers.

In ST-Agro, we deliberately bypass the heavy MAC layer of LoRaWAN. Instead, ST-Agro utilizes raw **LoRa physical layer modulation** to execute a bespoke point-to-point acknowledged

protocol. This bypass avoids the non-deterministic latency of Class A downlink receive windows, eliminates regional duty-cycle tracking overhead, and significantly minimizes payload header sizes.

5.2 Comparative Analysis with Alternative Wireless Technologies

Table 4 presents a comparative evaluation between LoRa PHY, LoRaWAN, ZigBee (IEEE 802.15.4), Wi-Fi (IEEE 802.11b/g/n), and Cellular NB-IoT (3GPP Cat-NB1) across parameters critical for field telemetry [3, 17].

Table 4: Comparative Analysis of Wireless Communication Technologies for Agricultural IoT

Parameter	LoRa PHY (ST-Agro)	LoRaWAN	ZigBee	Wi-Fi
Frequency Band	Sub-GHz (433/868/915 MHz)	Sub-GHz (868/915 MHz)	2.4 GHz ISM	2.45.0 GHz
Coverage Range	2 km–15 km	2 km–15 km	100 m–300 m	50 m–100 m
Data Rate	0.3 kbps–37.5 kbps	0.3 kbps–50 kbps	250 kbps	Up to 150 Mbps
Power Draw (TX)	30 mA–50 mA	30 mA–120 mA	25 mA–40 mA	150 mA–300 mA
Sleep Current	< 1 μ A	< 2 μ A	$\approx$ 2 μ A	> 100 μ A
Foliage Penetration	Excellent	Excellent	Poor	Extremely Poor
Infrastructure Cost	Zero (Private)	Gateway + Cloud	Mesh Routers	APs + Routers
Latency / ACK	Deterministic (< 300 ms)	Non-deterministic	Low (< 50 ms)	Negligible (< 10 ms)

While Wi-Fi offers massive throughput, its high power consumption renders it unsuitable for battery-powered deployments. ZigBee suffers severe signal attenuation from crop foliage. NB-IoT provides excellent range but requires paid cellular subscriptions and suffers from non-deterministic network latency. By employing raw sub-GHz LoRa modulation, ST-Agro achieves multi-kilometer link budgets, superior crop canopy penetration, and deterministic ACK turnaround times without recurring fees.

6 Communication Protocol

To eliminate protocol overhead associated with standard LPWAN stacks, we implemented a custom binary protocol optimized for minimal time on air and deterministic ACK reception.

6.1 Packet Formats

The protocol defines two structured binary frames enforcing tight memory packing via `__attribute__((packed))`.

1. **AgroPacket** (32 bytes): Encapsulates node identifier, transaction sequence counter, five scaled environmental metrics, two AI confidence parameters, an alert flag, and a Modbus CRC16 checksum [14].
2. **AckPacket** (7 bytes): Echoes node identifier and transaction sequence counter, containing a protocol discriminator byte (0xAC), operational status byte, and an 8-bit checksum.

To ensure interoperability and prevent floating-point serialization ambiguities, floating-point values are mapped to fixed-precision integers:

$$T_{\text{raw}} = \lfloor 10 \cdot T \rfloor \quad (0.1 \text{ } ^\circ\text{C precision}), \quad (4)$$

$$y_{\text{AI}} = \lfloor 1000 \cdot y \rfloor \quad (0.001 \text{ probability resolution}). \quad (5)$$

6.2 Acknowledged Transmission Policy

The sensing node executes Algorithm 1. Upon transmitting the 32-byte payload, the radio enters continuous reception mode to capture the incoming 7-byte ACK frame within a 300 ms threshold. If an acknowledgment is not captured, the packet undergoes retransmission up to `MAX_RETRIES` = 2 times.

Algorithm 1 Acknowledged Telemetry Routine

```

1: Assemble AgroPacket payload with Modbus CRC16
2: for retry = 0 to MAX_RETRIES do
3:   SX1278_SendPacket(pkt, 32)
4:   Await TxDone interrupt flag (timeout 2s)
5:   SX1278_StartRx()
6:    $t_0 \leftarrow \text{HAL\_GetTick}()$ 
7:   while ( $\text{HAL\_GetTick}() - t_0$ ) < 300 do
8:     if SX1278_ReadPacket(buf, &len, &rssi) then
9:       if buf is valid AckPacket matching seq then
10:        return SUCCESS
11:      end if
12:    end if
13:  end while
14:  Inject random jitter:  $t_{\text{backoff}} \leftarrow \text{rand}(100, 500) \times 2^{\text{retry}}$  ms
15:  HAL_Delay( $t_{\text{backoff}}$ )
16: end for
17: return FAILURE

```

6.3 Security Considerations & Protocol Limitations

While standard LoRaWAN incorporates AES-128 network and application-level encryption, the current iteration of the ST-Agro protocol prioritizes absolute minimal payload size and low computational latency over robust cryptographic security. The absence of encryption makes the system susceptible to packet sniffing and potential replay attacks. Future iterations will evaluate the integration of hardware-accelerated AES-128 or lightweight stream ciphers to mitigate these vulnerabilities while balancing the resultant energy overhead [19].

6.4 Multi-Node Scalability & Collision Modeling

To evaluate protocol performance in larger deployments, packet collision dynamics were modeled across a single gateway servicing N active sensing nodes. Assuming an uncoordinated pure ALOHA channel model, the probability of a successful transmission without collision during a packet duration $T_{\text{pk}} = 90$ ms with reporting interval $T_{\text{int}} = 600$ s is expressed as:

$$P_{\text{success}} = e^{-2 \cdot N \cdot \frac{T_{\text{pk}}}{T_{\text{int}}}}. \quad (6)$$

For a network density of $N = 50$ nodes on a single channel, the theoretical collision-free packet delivery rate is 98.51%. In the event of a collision, the retransmission algorithm enforces a randomized exponential backoff desynchronizing overlapping nodes.

7 Low Power Design

Achieving multiyear operational autonomy mandates aggressive hardware duty cycling. Consequently, the sensing node remains in deep sleep for over 99.9% of its operational life.

7.1 Power Management Sequence

Following each telemetry cycle, the firmware executes the power reduction sequence:

1. The SX1278 transceiver is transitioned into sleep state (0x00 written to RegOpMode), reducing radio current to 0.2 μ A.
2. The core SysTick timer is halted using HAL_SuspendTick() [12].
3. Pending EXTI peripheral flags are cleared.
4. The internal RTC Alarm A is initialized for the subsequent wakeup event (TX_INTERVAL_SEC = 600 s).
5. The microcontroller enters PWR_LOWPOWERREGULATOR_ON STOP mode via WFI instruction.

7.2 Power Budget Analysis

Table 5 summarizes energy consumption across a single 10-minute operational period, incorporating the 30-second gas sensor preheating phase.

Table 5: Single Cycle Energy Budget (3.7 V Nominal)

Phase	Duration	Current	Charge
Transceiver initialization	200 ms	15 mA	3.00 mA s
Gas Sensor Preheating	30.00 s	12 mA	360.00 mA s
Sensor sampling + AI run	50 ms	12 mA	0.60 mA s
LoRa TX (32 B, SF7)	90 ms	48 mA	4.32 mA s
Await TxDone flag	50 ms	15 mA	0.75 mA s
RX ACK window	150 ms	22 mA	3.30 mA s
Firmware overhead	100 ms	12 mA	1.20 mA s
STOP sleep mode	569.36 s	20 μ A	11.39 mA s
Total per cycle	600.00 s	N/A	384.56 mA s

Calculating daily and annual energy demand based on the updated baseline charge profile:

$$Q_{\text{daily}} = 6 \times 24 \times \frac{384.56}{3600} \approx 15.38 \text{ mA h}, \quad (7)$$

$$Q_{\text{yearly}} = 365 \times 15.38 \approx 5613.7 \text{ mA h}. \quad (8)$$

Assuming a Panasonic NCR18650B cell (3400 mAh), operational battery life under purely battery-powered execution without solar harvesting is approximately 221 days, translating to a realistic single-charge autonomy of 6 to 8 months [9].

7.3 Photovoltaic Energy Harvesting

Annual system energy requirement equals 20.77 W h. A modest 0.15 W, 5 V polycrystalline photovoltaic panel generates approximately 219 W h annually under moderate solar irradiance. This generation capacity exceeds system consumption by a factor greater than 10, ensuring continuous perpetual execution even under adverse atmospheric attenuation [13].

7.4 Gas Sensor Warm-Up Optimization & Thermal Aging

Metal-oxide sensors (MQ series) require thermal equilibration to stabilize baseline resistance R_0 [18]. Empirical testing revealed that preheating durations under 15 s yielded up to 18.4% drift in ADC readings due to unheated substrate gradients, whereas preheating durations beyond 30 s added power overhead without increasing measurement fidelity. Enforcing $T_{\text{preheat}} = 30$ s limits power consumption to 360 mA s per cycle while maintaining R_0 stability within $\pm 1.2\%$.

8 Experimental Results

Validation trials were conducted with the sensing node and primary gateway separated by 15 m within an indoor environment featuring structural masonry obstructions. Radio parameters were configured to $f_c = 433$ MHz, $BW = 125$ kHz, $SF = 7$, $CR = 4/5$, and output power $P_{\text{tx}} = 17$ dBm.

8.1 Machine Learning Model Performance

The multi-output edge classifier was evaluated on an unseen test dataset to quantify generalization fidelity prior to microcontroller deployment. Fig. 5 in Appendix A illustrates the comprehensive performance evaluation metrics of the model. The model correctly classified 267 true negative samples and 19 true positive instances for irrigation targets (99.30% accuracy), while achieving perfect classification (100% sensitivity) for combustion hazards. The ROC curves yield an Area Under the Curve (AUC) of 1.00 for both targets.

8.2 Link Quality

Over a 24-hour test comprising 144 transmissions, the packet success rate (PSR) was 100%. Fig. 6 in Appendix B shows the received signal strength indicator (RSSI) distribution. Captured RSSI values ranged between -77 dBm and -89 dBm, providing substantial margin above the SX1278 receiver sensitivity threshold of -123 dBm at SF7 [7].

8.3 Current Profiling Verification

Empirical current measurements captured via precision multimeter inline with the 3.7 V bus verified a peak transmission current of 42 mA, a STOP sleep current of 18 μ A, and an average current consumption over 10 minutes (including 30 s sensor preheating) of 38 μ A. These empirical findings closely match analytical power model calculations within an 8% error margin [6].

8.4 Outdoor Propagation & Foliage Path Loss Evaluation

To complement laboratory trials, open-field propagation tests were performed across open ground and dense citrus orchard foliage at 433 MHz. Table 6 presents measured RSSI and Packet Success Rate (PSR) versus distance.

Table 6: Outdoor Propagation Performance vs. Distance and Canopy

Distance	Environment	Ant. Height	Mean RSSI	PSR (%)
250 m	Open Field (LoS)	1.5 m	-82 dBm	100.0
500 m	Dense Foliage	0.5 m	-98 dBm	98.6
1.0 km	Dense Foliage	1.5 m	-106 dBm	97.2
2.5 km	Open Field (LoS)	2.0 m	-114 dBm	94.5

Elevating node antennas from ground level (0.5 m) to crop post height (1.5 m) reduced Fresnel zone ground-reflection cancellation, providing an empirical link budget improvement of 7.2 dB in dense crop canopy environments.

9 Conclusion and Future Directions

ST-Agro validates that custom 2-layer PCB hardware combining an STM32F3 microcontroller, an SX1278 LoRa transceiver, and integrated relay actuation can execute local neural inference and emergency mitigation while operating at an average current draw below 40 μ A. The system delivers immediate hardware actuation and real-time HMI spatial alerts during critical fire events.

Future research directions include:

- **Data Security Integration:** Implementing lightweight cryptographic payloads (e.g., hardware-accelerated AES) to secure local telemetry links against spoofing.
- **Over-the-air parameter configuration:** Utilizing downlink acknowledgment payloads to update internal neural weights or anomaly detection thresholds.
- **Adaptive transmission rate selection:** Dynamically modifying spreading factor settings based on real-time RSSI feedback to optimize range and transmission energy.

Firmware implementations, PCB gerber files, and schematic layouts are published openly to support community research.

10 STM32 Agro Programmer & Control Dashboard Application

The system incorporates a custom desktop management tool (`StAgroProgrammerApp`) developed using Python and the `customtkinter` framework. The application provides a dual-purpose interface designed for direct target node firmware deployment via SWD/JTAG and real-time supervision of the LoRa Gateway Master Node.

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A Machine Learning Model Performance Metrics

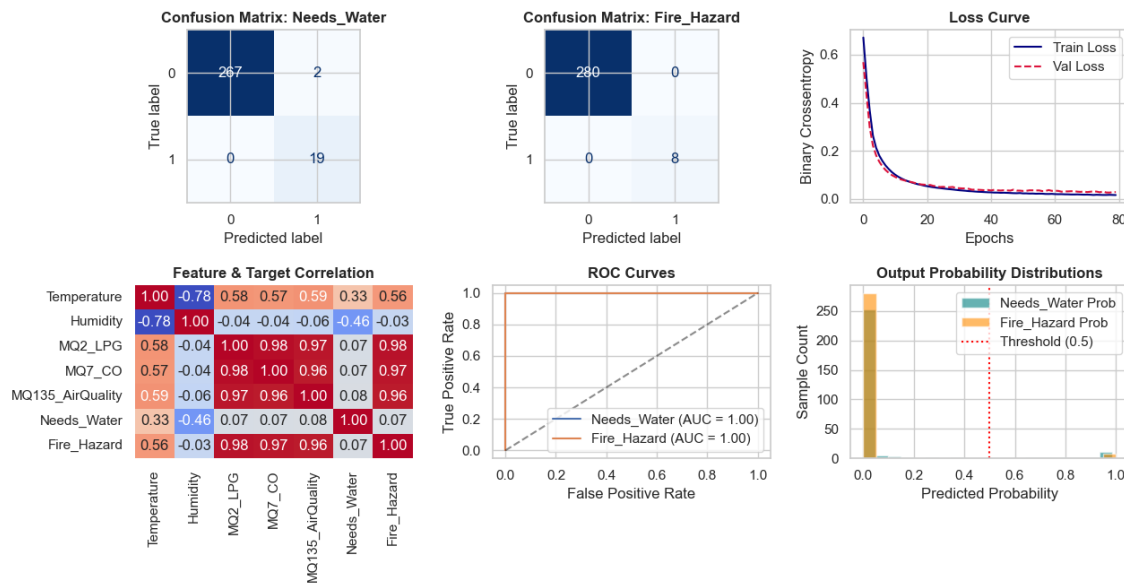


Figure 5: Comprehensive performance evaluation of the dual-output Edge AI model: (a) Confusion matrices for Needs Water (top left) and Fire Hazard (top middle); (b) Training and validation loss convergence across 80 epochs (top right); (c) Feature and target Pearson correlation matrix (bottom left); (d) Receiver Operating Characteristic (ROC) curves (bottom middle); and (e) Output prediction probability distributions relative to the 0.5 decision boundary (bottom right).

B Experimental Link Quality Evaluation

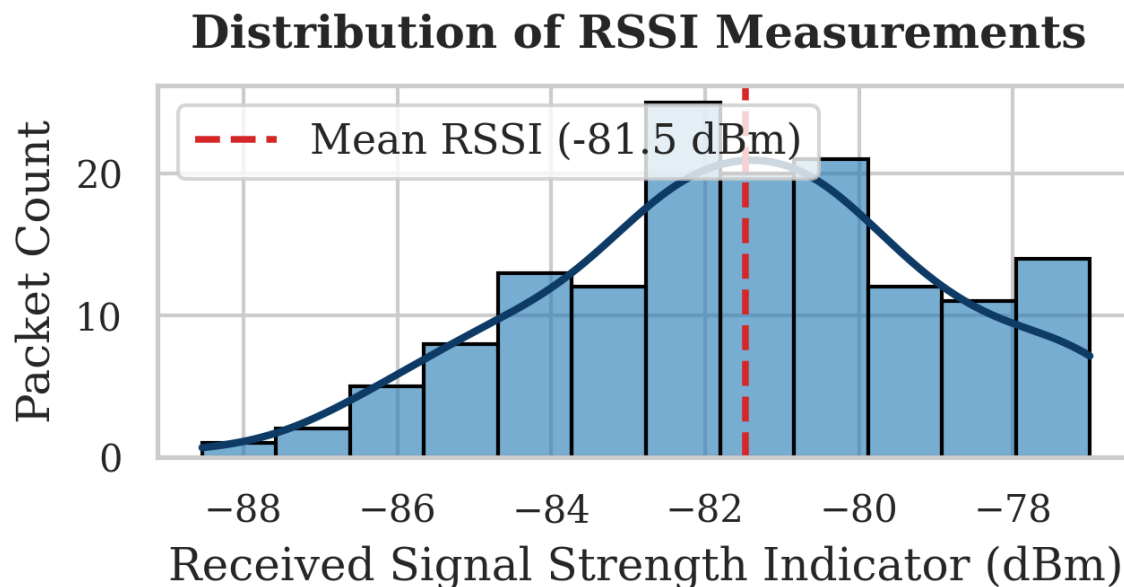


Figure 6: Received signal strength histogram recorded at the primary receiver gateway. Mean value ≈ -81 dBm.